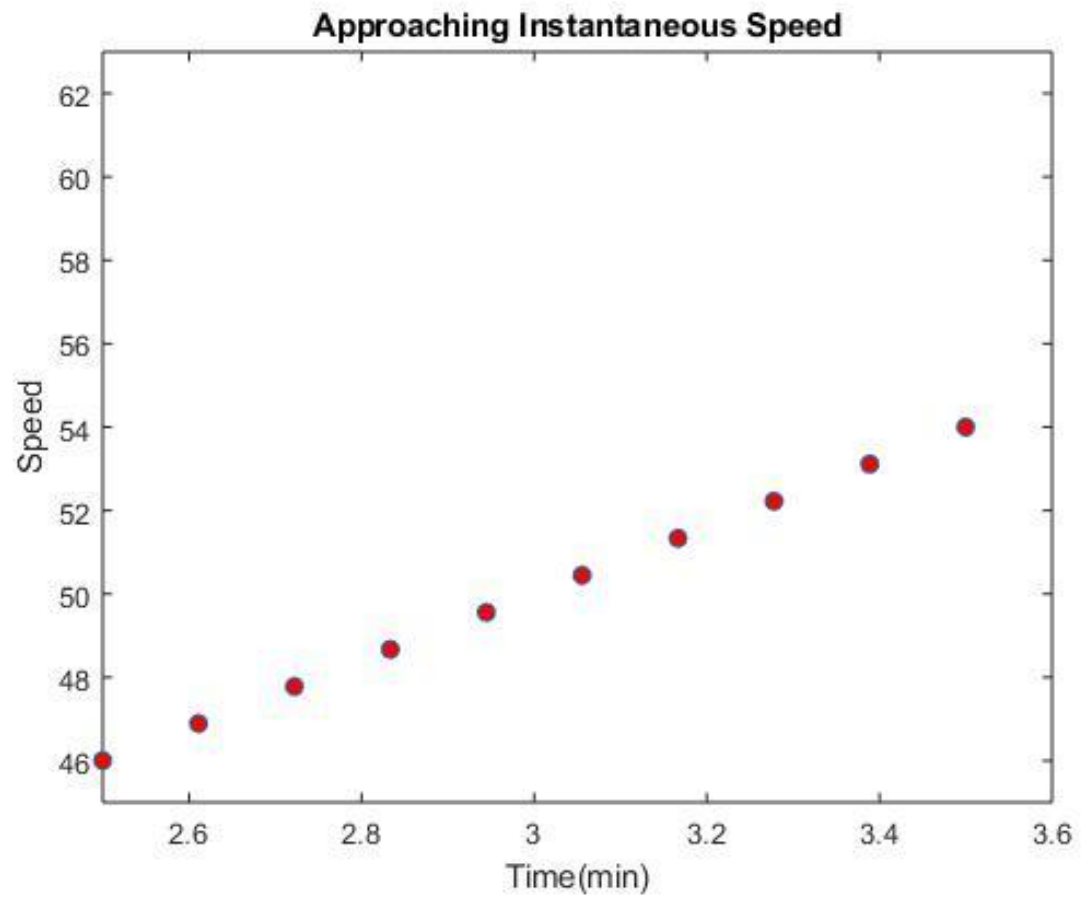


Limits, Continuity and Differentiability

Afifa Tabassum Chaity

Limit

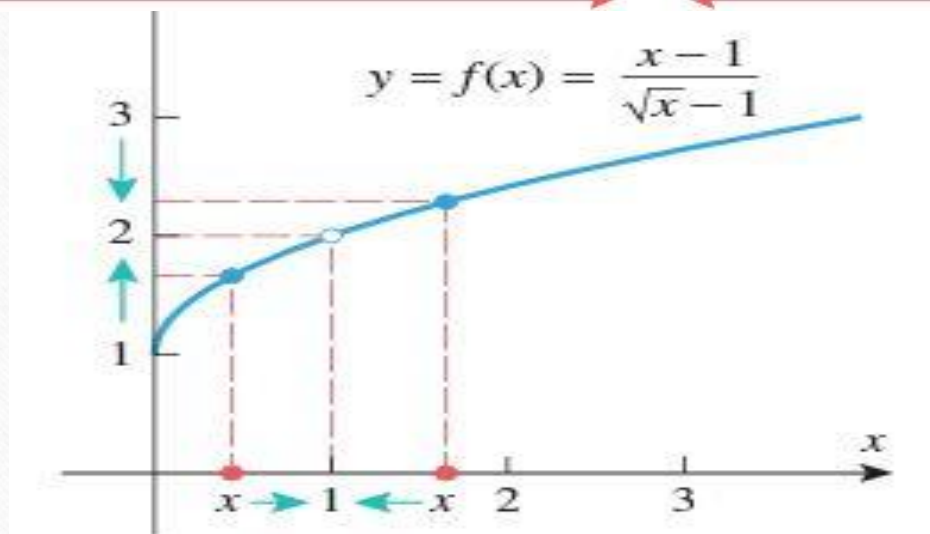


Limit

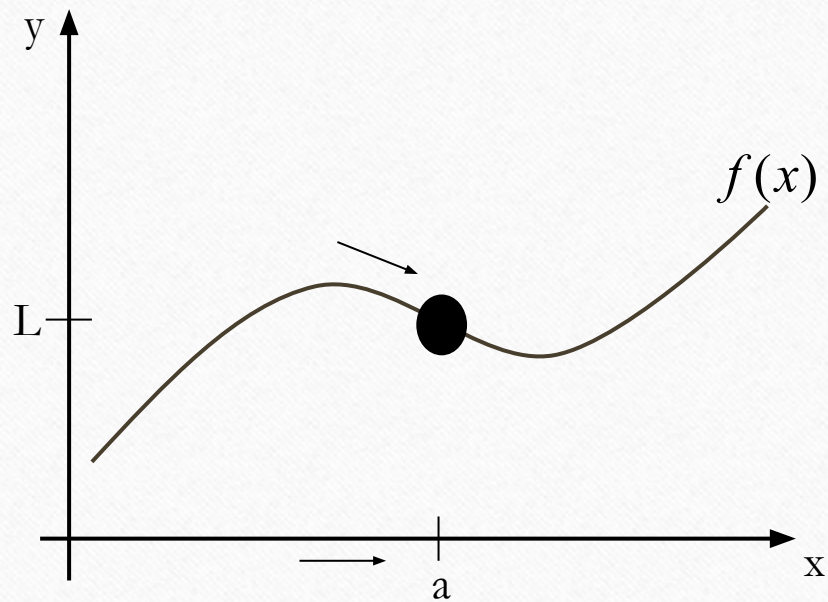
Use numerical evidence to make a conjecture about the value of

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$$

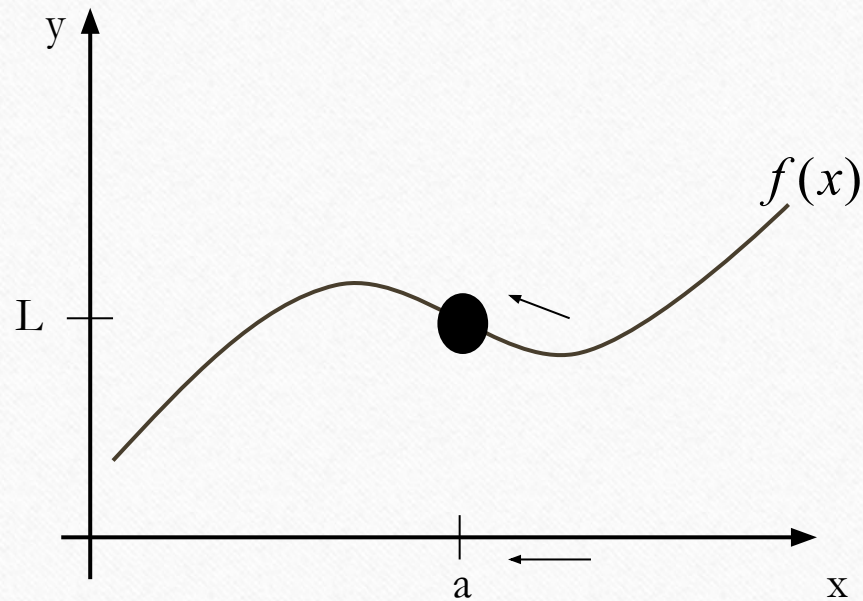
x	0.99	0.999	0.9999	0.99999		1.00001	1.0001	1.001	1.01
$f(x)$	1.994987	1.999500	1.999950	1.999995		2.000005	2.000050	2.000500	2.004988



One sided limits

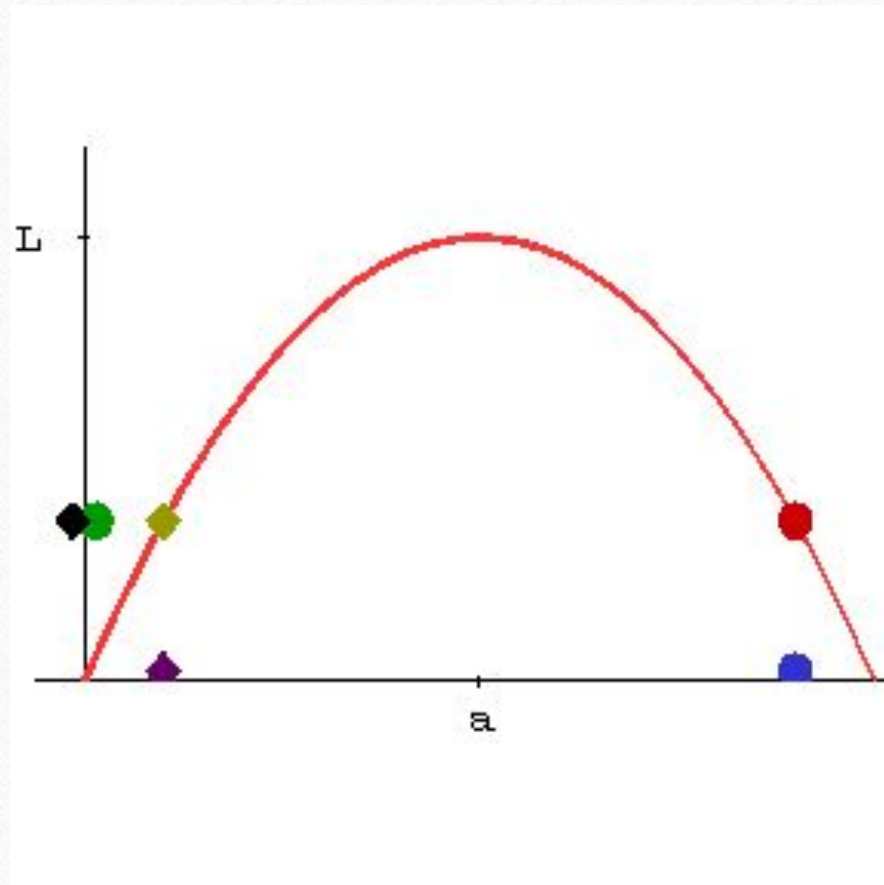


$$\lim_{x \rightarrow a^-} f(x) = \text{Left sided limit}$$

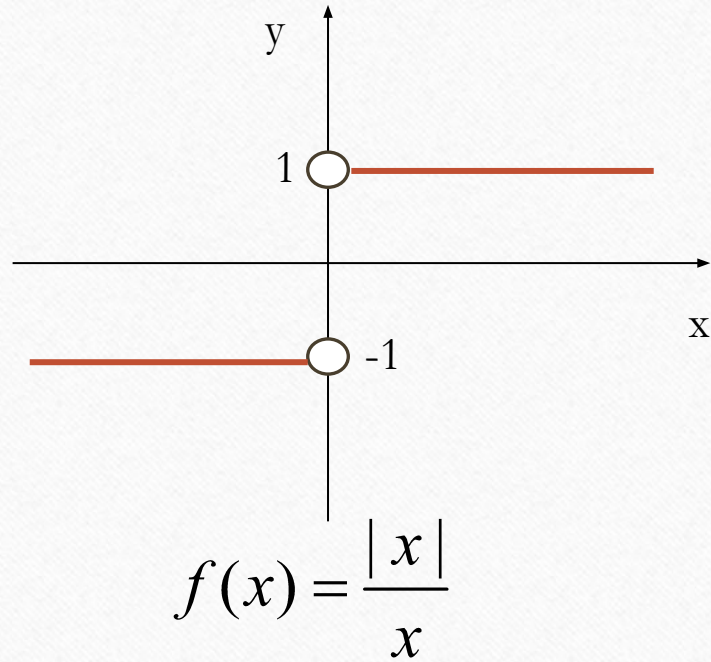


$$\lim_{x \rightarrow a^+} f(x) = \text{Right sided limit}$$

One sided limits

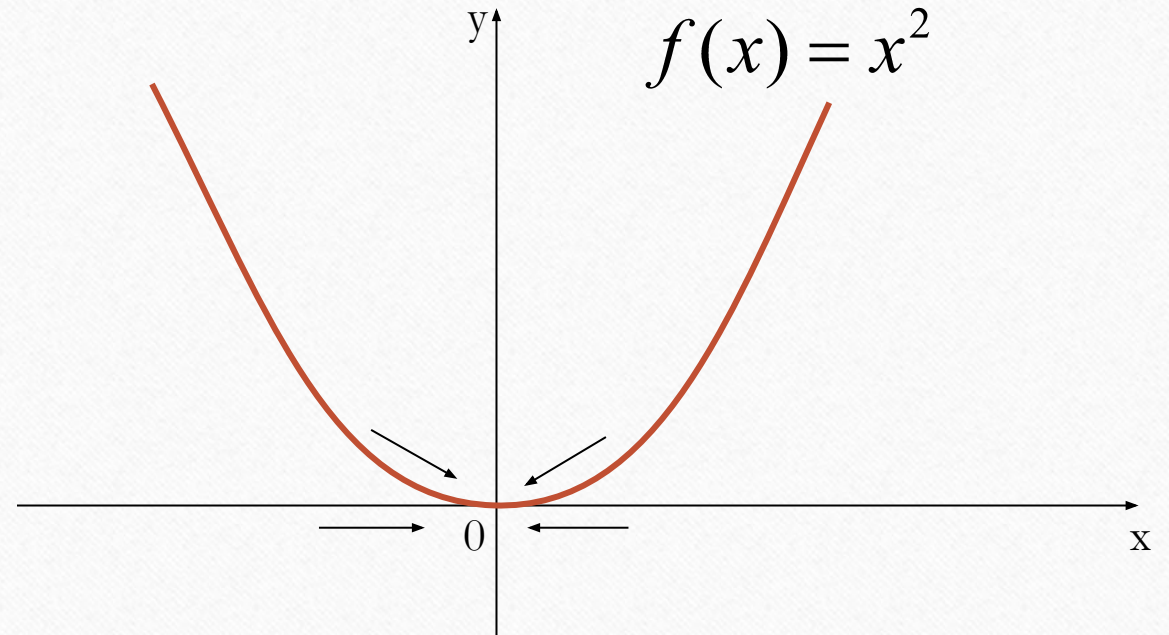


Existence of limit



$$\lim_{x \rightarrow 0^-} f(x) = -1 \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = 1$$

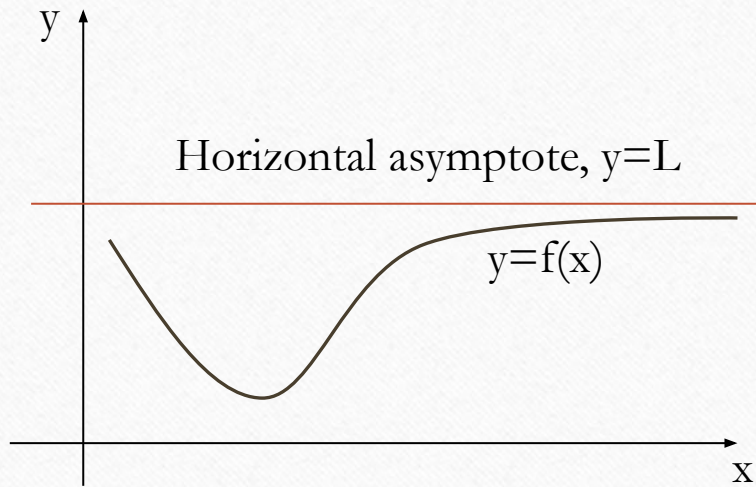
Limit doesn't exist



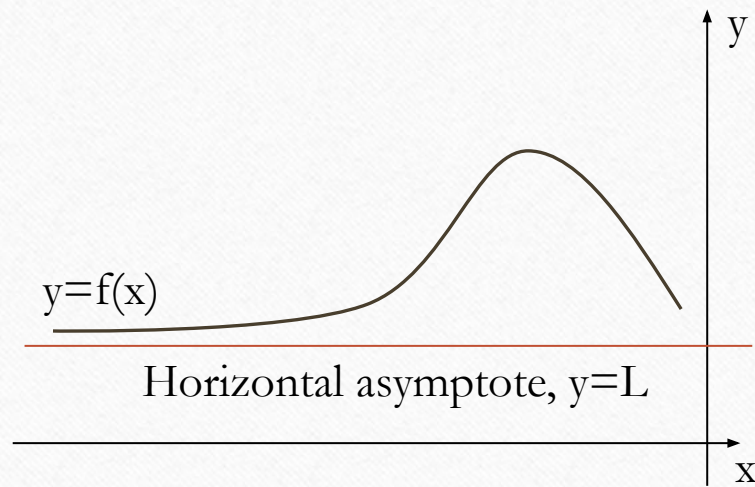
$$\lim_{x \rightarrow 0^-} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} f(x) = 0$$

Limit exists

Limits at infinity



$$\lim_{x \rightarrow +\infty} f(x) = L$$



$$\lim_{x \rightarrow -\infty} f(x) = L$$

Properties of limit

Let, a be a real number and

$$\lim_{x \rightarrow a} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = L_2$$

Then:

$$1. \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L_1 + L_2$$

$$4. \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2}$$

$$2. \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L_1 - L_2$$

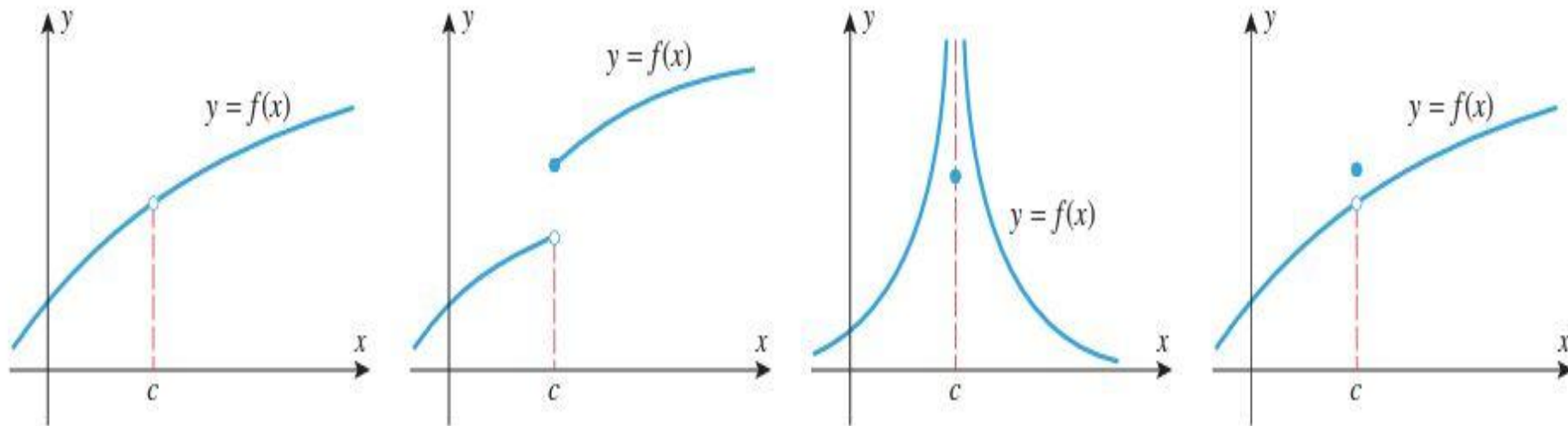
$$5. \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} = \sqrt[n]{L_1}$$

$$3. \lim_{x \rightarrow a} [f(x)g(x)] = [\lim_{x \rightarrow a} f(x)][\lim_{x \rightarrow a} g(x)] = L_1 L_2$$

Applications of Limit

- Studying the behavior of a function.
- Checking the continuity of a function
- Analysis of market behavior.
- Analysis of response of body to various factors over time.
- The maximum load a bridge can bear is determined by considering the limits of material strength and structural integrity.
- Analysis of changes in ecosystems over time.

Continuity

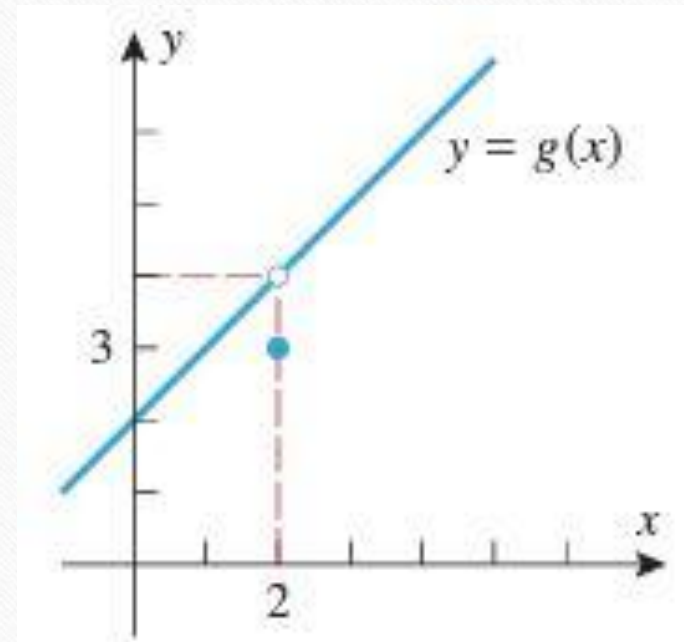


Applications of continuity

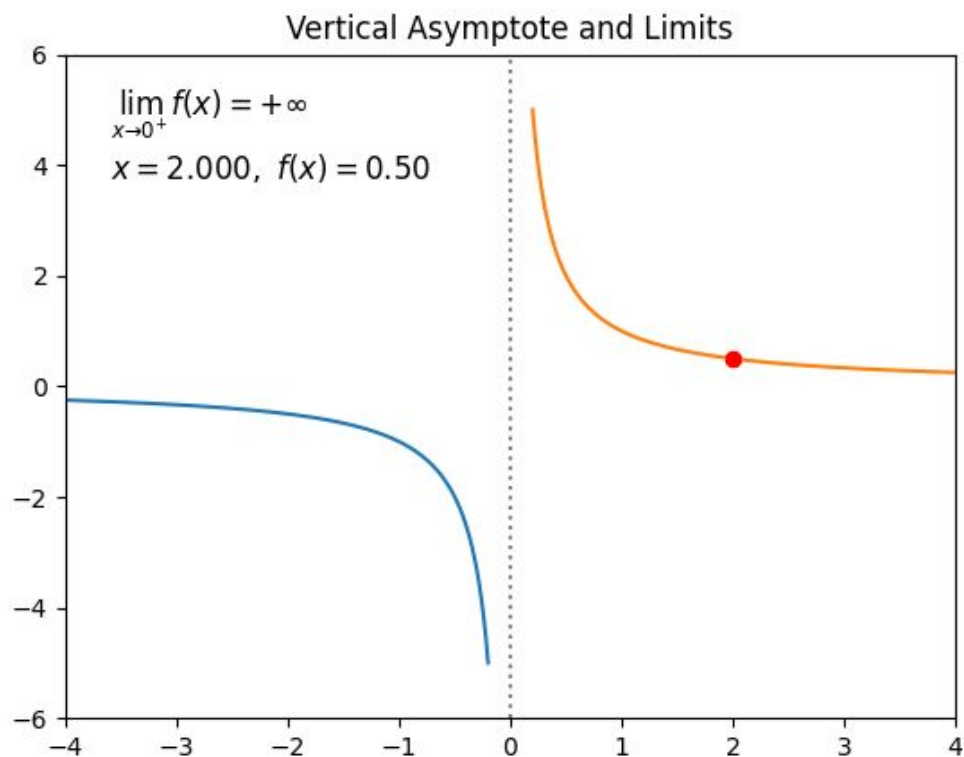
- Determine if the function is continuous at $x=2$

$$g(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & x \neq 2 \\ 3, & x = 2, \end{cases}$$

$$\lim_{x \rightarrow 2} g(x) \neq \lim_{x \rightarrow 2} (x + 2) = 4$$

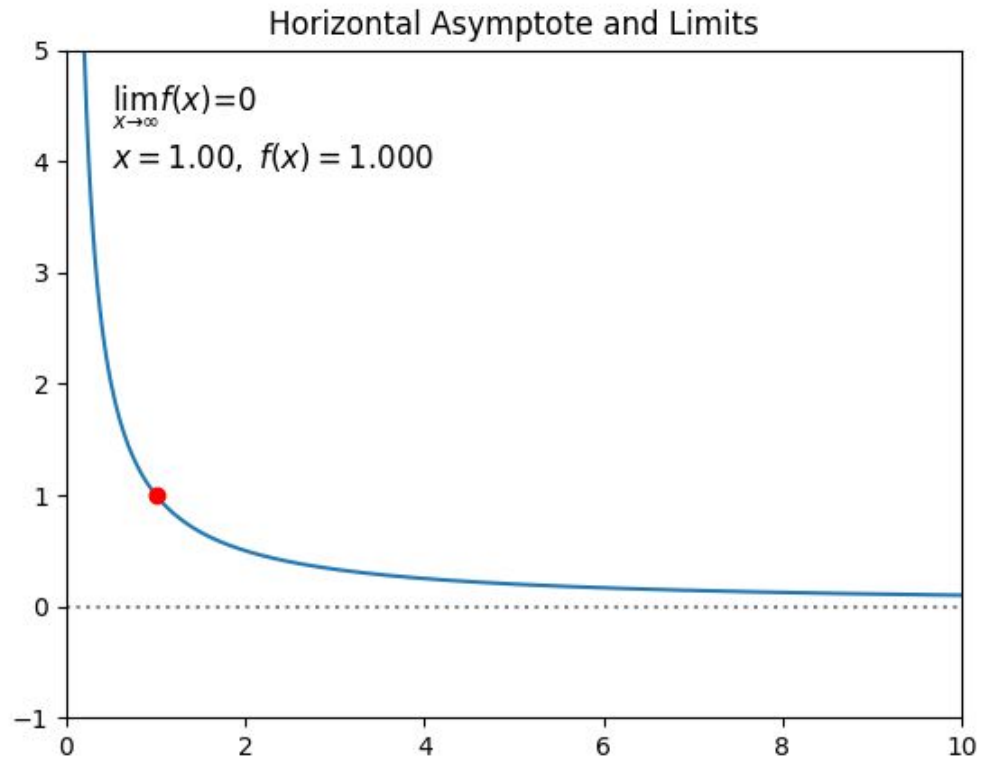


Vertical Asymptote



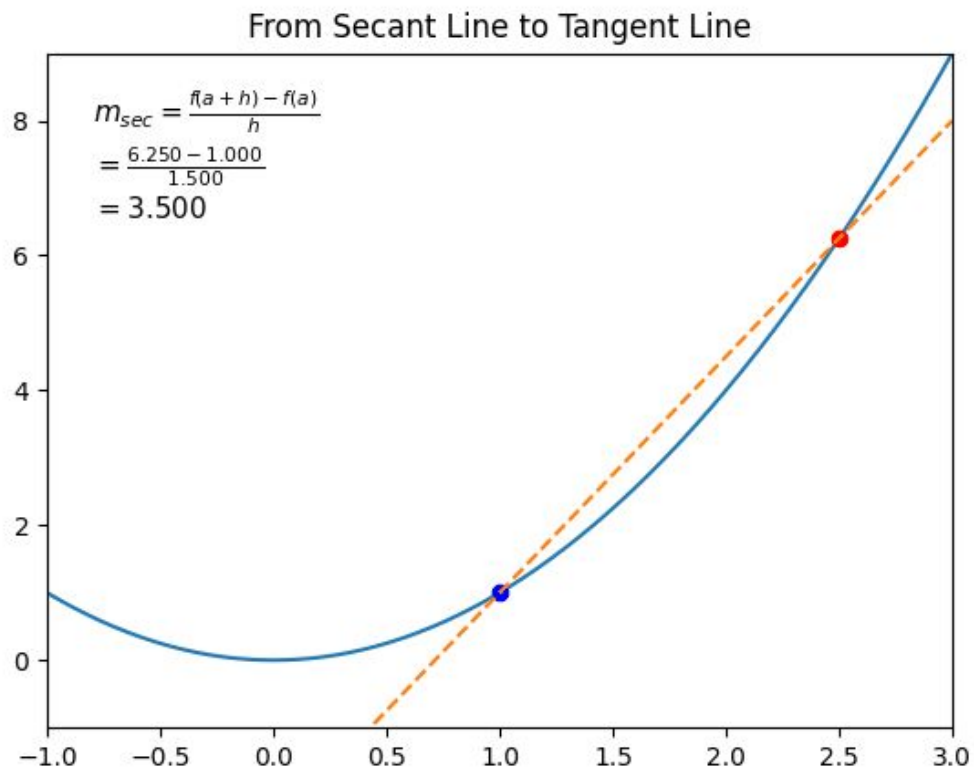
$$\lim_{x \rightarrow a} f(x) = \infty$$

Horizontal Asymptote



$$\lim_{x \rightarrow \infty} f(x) = L$$

Differentiability



$$m_{sec} = \frac{f(x+a) - f(x)}{(x+a) - x}$$

$$m_{sec} = \lim_{a \rightarrow x} \frac{f(x+a) - f(x)}{(x+a) - x}$$

$$m_{tan} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Differentiability

A function f is called differentiable at a point x_0 if the limit

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \dots\dots\dots(1)$$

exists. (1) is called the derivative function of $f(x)$.

If f is differentiable in $(-\infty, \infty)$, then f is differentiable everywhere.

Left Hand Derivative

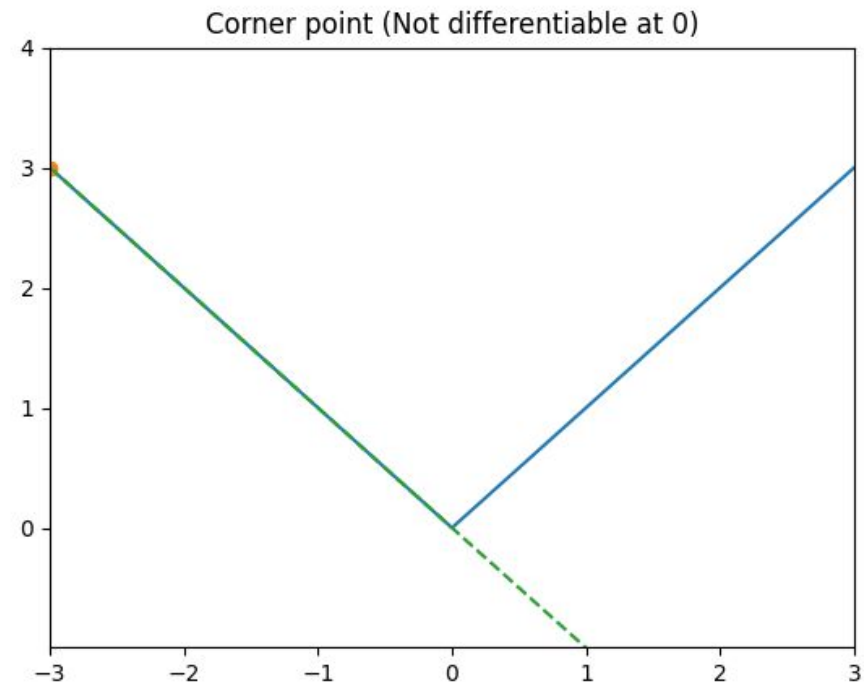
$$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$$

Right Hand Derivative

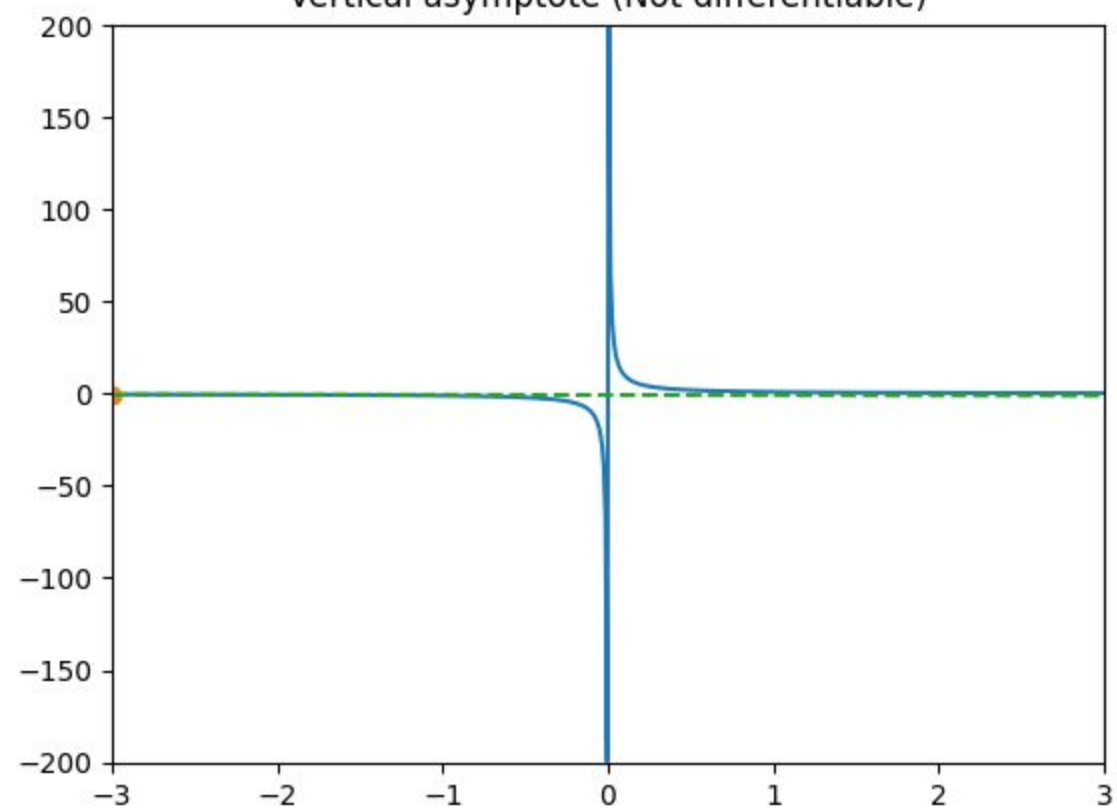
$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

A function f is not differentiable if in the graph of f there's

- Corner points
- Points of vertical tangency exists.



Vertical asymptote (Not differentiable)



1. Check the continuity and differentiability at $x=0$

$$f(x) = \begin{cases} \frac{e^{\frac{1}{x^2}}}{e^{\frac{1}{x^2}} - 1} & , x \neq 0 \\ 1 & , x = 0 \end{cases}$$

Solution:

Continuity at $x=0$:

1. $f(0)=1$

2. L.H.L. = $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x^2}}}{e^{\frac{1}{x^2}} - 1}$

$$= \lim_{x \rightarrow 0^-} \frac{e^{\frac{1}{x^2}}}{e^{\frac{1}{x^2}}(1 - e^{-\frac{1}{x^2}})} =$$
$$\lim_{x \rightarrow 0^-} \frac{1}{(1 - e^{-\frac{1}{x^2}})} = \frac{1}{(1 - e^{-\infty})} = \frac{1}{(1 - 0)} = 1$$

$$\text{R.H.L.} = \lim_{x \rightarrow 0^+} \frac{e^{\frac{1}{x^2}}}{e^{\frac{1}{x^2}}(1 - e^{-\frac{1}{x^2}})} =$$
$$\lim_{x \rightarrow 0^+} \frac{1}{(1 - e^{-\frac{1}{x^2}})} = \frac{1}{(1 - e^{-\infty})} = \frac{1}{(1 - 0)} = 1$$

$$\text{L.H.L.} = \text{R.H.L.}$$

$\therefore \lim_{x \rightarrow 0} f(x)$ exists

3. $\lim_{x \rightarrow 0} f(x) = 1 = f(0)$

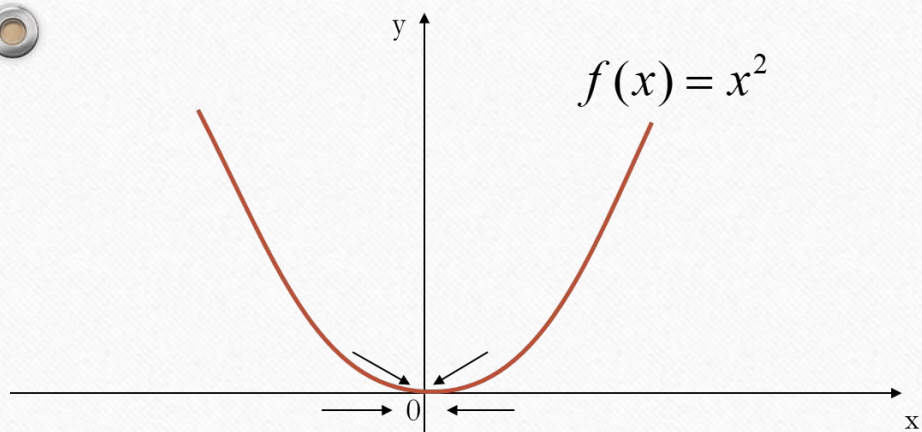
So, $f(x)$ is continuous at $x=0$

Differentiability at $x=0$:

$$f'(0)$$

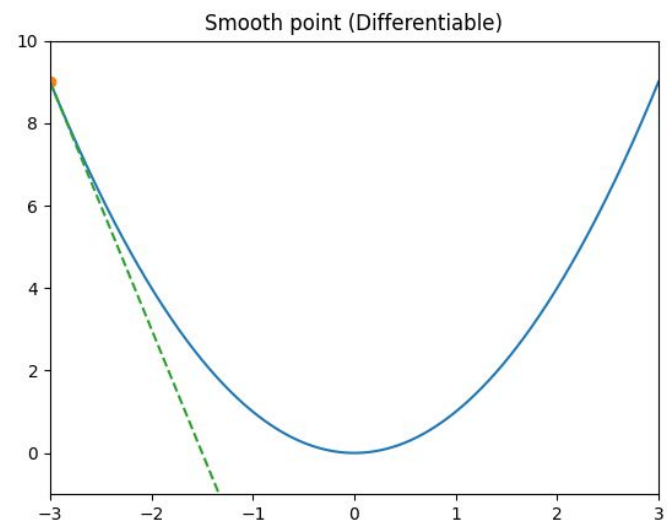
$$= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = \lim_{h \rightarrow 0} \frac{\frac{e^{\frac{1}{h^2}}}{e^{\frac{1}{h^2}} - 1} - 1}{h} = \lim_{h \rightarrow 0} \frac{\frac{e^{\frac{1}{h^2}} - e^{\frac{1}{h^2}} + 1}{e^{\frac{1}{h^2}} - 1}}{h} = \lim_{h \rightarrow 0} \frac{1}{h(1 - e^{\frac{1}{h^2}})} = \frac{1}{\infty} = 0$$

$$f'(0)=0$$



Differentiable function

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = 2x_0$$



Appendix

Applications of differentiation

- Rate of change of function with time (velocity, acceleration)
- Finding the maximum or minimum of a function, and identify critical points where the derivative is zero
- Measuring population growth, radioactive decay, and the spread of diseases.

1.4.1 LIMIT DEFINITION Let $f(x)$ be defined for all x in some open interval containing the number a , with the possible exception that $f(x)$ need not be defined at a . We will write

$$\lim_{x \rightarrow a} f(x) = L$$

if given any number $\epsilon > 0$ we can find a number $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{if} \quad 0 < |x - a| < \delta$$

Reference

- Calculus-Early Transcendentals, Howard Anton, 10th edition, 2017, pages 67-150
- https://phys23p.sl.psu.edu/~mrg3/mathanim/calc_I/1_sided_limit_matched.html
https://phys23p.sl.psu.edu/~mrg3/mathanim/calc_I/1_sided_limit_matched.html